Are Space and Time Fundamental Concepts?

The German philosopher Immanuel Kant suggested that it would be not merely difficult to do away with space and time when thinking about and describing the universe, it would be downright impossible. Frankly, I can see where Kant was coming from. Whenever I sit, close my eyes, and try to think about things while somehow not depicting them as occupying space or experiencing the passage of time, I fall short. Way short. Space, through context, or time, through change, always manages to seep in. Ironically, the closest I come to ridding my thoughts of a direct spacetime association is when I'm immersed in a mathematical calculation (often having to do with spacetime!), because the nature of the exercise seems able to engulf my thoughts, if only momentarily, in an abstract setting that seems devoid of space and time. But the thoughts themselves and the body in which they take place are, all the same, very much part of familiar space and time. Truly eluding space and time makes escaping your shadow a cakewalk.

Nevertheless, many of today's leading physicists suspect that space and time, although pervasive, may not be truly fundamental. Just as the hardness of a cannonball emerges from the collective properties of its atoms, and just as the smell of a rose emerges from the collective properties of its molecules, and just as the swiftness of a cheetah emerges from the collective properties of its muscles, nerves, and bones, so too, the properties of space and time—our preoccupation for much of this book— may also emerge from the collective behavior of some other, more fundamental constituents, which we've yet to identify.

Physicists sometimes sum up this possibility by saying that spacetime may be an illusion—a provocative depiction, but one whose meaning requires proper interpretation. After all, if you were to be hit by a speeding cannonball, or inhale the alluring fragrance of a rose, or catch sight of a blisteringly fast cheetah, you wouldn't deny their existence simply because each is composed of finer, more basic entities. To the contrary, I think most of us would agree that these agglomerations of matter exist, and moreover, that there is much to be learned from studying how their familiar characteristics emerge from their atomic constituents. But because they are composites, what we wouldn't try to do is build a theory of the universe based on cannonballs, roses, or